

“Autonomous Water Surface Cleaning System”

*“Water bodies are silently **drowning beneath** floating waste.
What if **cleanup could** happen autonomously?”*

100%

Edge Autonomous

0

Moving Collection Parts

AI/ML

Edge Debris Tracking

Debris like PET bottles, cups, and styrofoam sheets blocks stagnant water surfaces, preventing oxygen diffusion and suffocating ecosystems. Hazardous Manual Cleaning: Human intervention in drainage channels, temple tanks, and marshy lakes is highly unsafe, unhygienic, and heavily labor-intensive. Inefficient Economics: Traditional waste cleanup relies on costly heavy machinery or continuous labor deployments with zero persistent results.

URBAN PONDS & TEMPLE TANKS

Debris Stagnation

Unmanaged plastics block municipal spillways, temple tanks, and narrow park ponds, resulting in stagnant water, odors, and breeding grounds for vectors.

Standard commercial water skimmers are too large or costly for shallow municipal water bodies.

Thick weed mats wrap around rotating gears, locking motors and burning coils.
Small-Scale Isolation: Local water ponds are physically inaccessible to standard industrial harbor boats.

Industrial Scale Only

Heavy trash skimmers are too large and heavy to fit or navigate narrow canals, shallow residential ponds, or historical temple tanks.

High Mechanical Jams

Traditional conveyor belts and rotating drums constantly seize when weeds, water hyacinths, or plastics wind into wheels.

Cloud Dependency

Smart drones rely heavily on continuous cellular or cloud data links, failing under bridge passages, tree covers, or remote ponds.

Prohibitive Capital

High acquisition costs coupled with continuous diesel fuel and maintenance overheads lock out local authorities.

Hydrodynamically optimized double catamaran fiberglass hulls provide extreme stability, high buoyancy, and minimal drag.

Zero Moving Part Collection: A passive catamaran induction channel funnels debris directly into an internal mesh net. No conveyor belts. Zero weed jams.

Onboard Edge AI Processing: Features a local Raspberry Pi running computer vision and target tracking models entirely offline without cellular links.

Low Cost & High Autonomy: Fully battery-powered differential marine thruster navigation achieves up to 4 hours of autonomous sweeping.

A diagram showing a cross-section of a passive induction flow channel. It consists of two vertical blue bars on a dark background, representing the channel walls. The text 'PASSIVE INDUCTION FLOW CHANNEL' is centered between the bars.

**PASSIVE INDUCTION
FLOW CHANNEL**

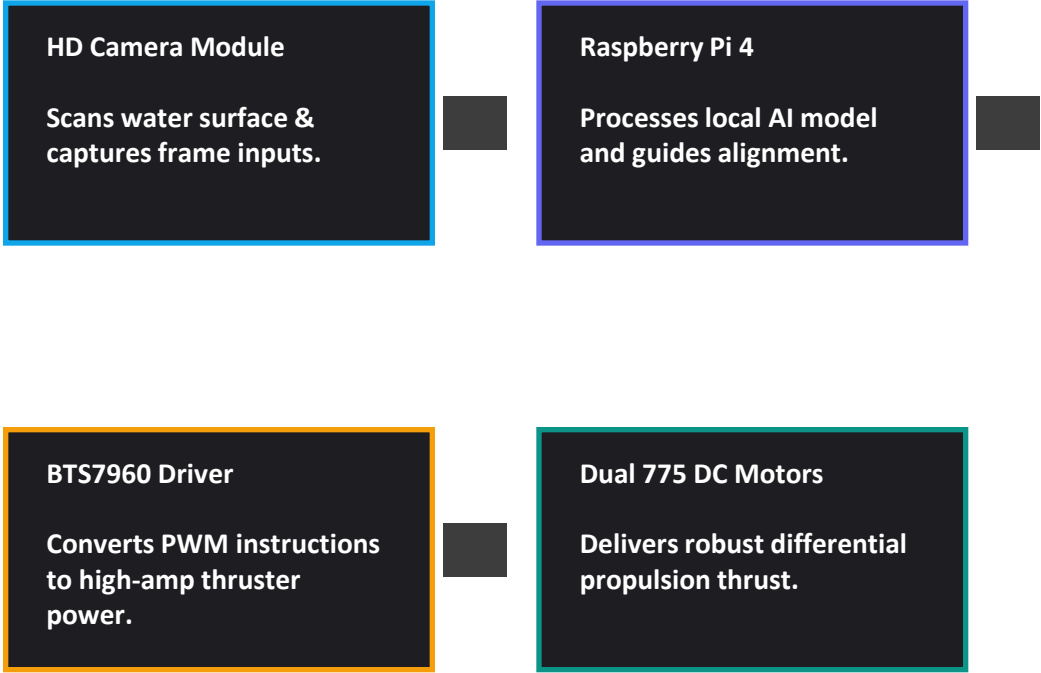
[Zero Moving Parts]

Captures a 90-degree field-of-view of the water surface, feeding live JPEG frames directly to the edge processor.

Raspberry Pi Node: The computing brain. Executes real-time MobileNetV2-SSD local inferences, differential steering, and navigation loops.

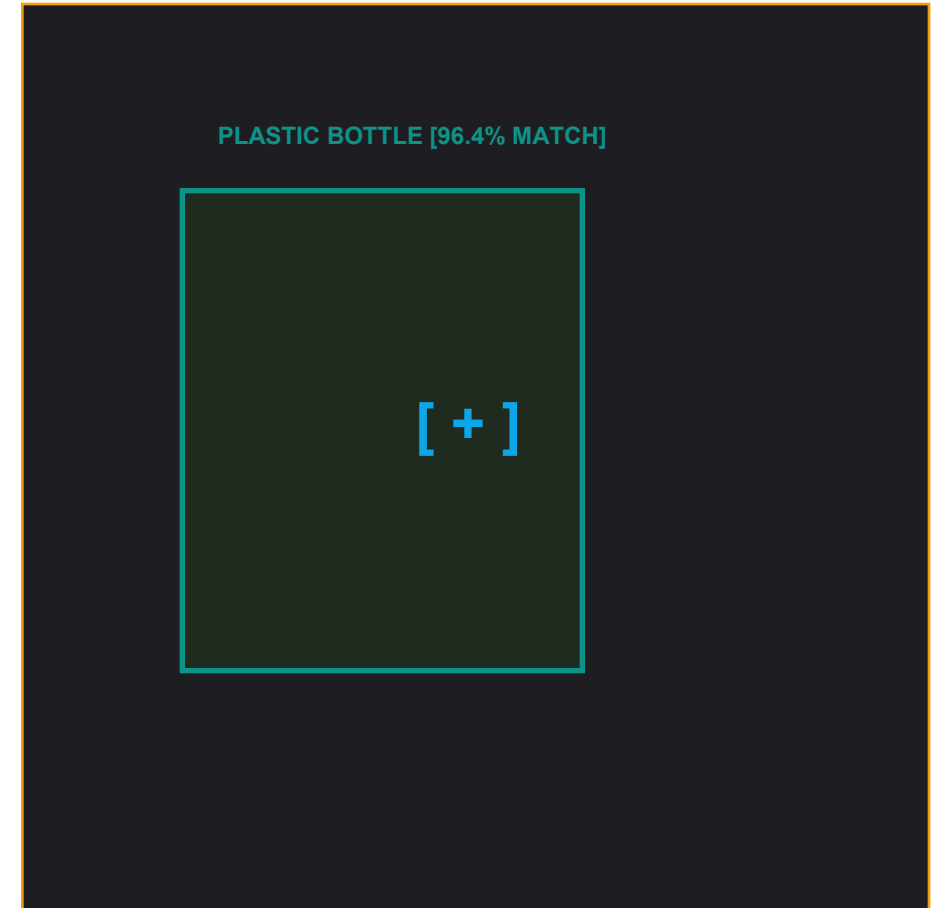
PWM Motor Driver: Commands high-current BTS7960/L298N drivers to dial-in precise differential thruster speeds for target centering.

Dual-Rail Power System: Onboard 12V LiFePO4 battery pack powers propulsion directly, stepped down to a 5V buck rail for the processor.



Front HD camera scans the water surface in real-time, detecting PET bottles, plastic cups, or styrofoam debris.

2. Dynamic Lock & Align: The tracking loop locks onto targets and aligns the catamaran by adjusting differential thruster power outputs.
3. Passive Capture Induction: Catamaran drives over the debris. The central induction channel funnels it into the mesh net, letting water pass.
4. Telemetry Update: Edge computer reports path progress, debris metrics, net volume capacity, and motor loads back to the command dashboard.



Twin hull displacement: $V = 2 * (L * W * D) = 2 * (0.7m * 0.1m * 0.12m) = 16.8$ Liters. Supports a 6.0 kg vessel dry mass and 4.0 kg debris payload at 59.5% hull draft height.

Propulsion & Hydrodynamics: Dual 775 high-torque DC motors driven via differential PWM, drawing 3A average. Onboard 12V 10Ah LiFePO4 battery provides ~3.3 hours of runtime at 0.7 m/s speed.

Edge Inference Performance: Quantized MobileNetV2-SSD running locally via TensorFlow Lite on Raspberry Pi 4 CPU. Runs at ~10 FPS, matching differential navigation loop requirements.

BUOYANCY EQUILIBRIUM

$$F_b = p_{water} * V_{disp} * g$$

Twin Hull Volume (Max): 16.8 Liters

Dry Mass of Vessel: 6.0 kg
Mesh Debris Payload: 4.0 kg

Total Displaced Mass: 10.0 kg
Designed Hull Draft: 59.5%

Cruising Speed: 0.7 m/s
Average Battery Current: 3.0 A

Metric	EcoFloat V2	Industrial Skimmers	Manual Labor
Acquisition Cost	Ultra Low (~)	High (,000+)	Wages (/yr)
Moving Parts Jams	Zero (Passive Duct)	High (Conveyors)	None
AI Debris Tracking	Yes (Local Edge)	None	None
Small Water Suitability	Excellent (Shallow)	Poor (Too bulky)	Good
Telemetry Panel	Yes (Real-time)	None	None
Human Safety Risk	Zero Risk (Autonomous)	Low (Driver boat)	High (Hazardous)

Catching PET and Styrofoam blocks before solar UV radiation breaks them down into permanent, toxic microplastics that enter the local food chain.

Dissolved Oxygen Restoration: Removing surface film waste allows natural atmospheric gas exchange, drastically improving water health and raising dissolved oxygen (DO) profiles.

Safe Habitats: Reclaims pristine habitats for local bird species, fish, and aquatic plants in historical urban temple tanks and neighborhood community ponds.

Pond Surface Clearing
Drastically increases sunlight penetration and aesthetic value of urban ponds.

Operational Safety
Eliminates the need for manual sorting or wading in toxic urban runoff.

Preventative Interception
Reclaims floating waste before solar UV rays decompose it into microplastics.

Integrating flexible photovoltaic panels on top of the catamaran hulls, extending operational autonomy to up to 8 hours daily during sunlight.

IoT Swarm Fleet Dynamics: Utilizing local radio modules, multiple EcoFloat units can coordinate cleaning sweeps collectively as a swarm, mapping large lakes in hours.

Smart City Environmental SaaS: Hooking edge data back into municipal control networks, providing localized water health mapping, heatmaps, and plastic volume forecasts.



SWARM COOPERATIVE CONTROL

Units divide sweeping areas into grids, sharing GPS tracks via long-range radio (LoRa) links.

- Program perimeter GPS waypoints around the stagnant body.
2. Launch & Auto-Sweep: Turn key autonomous glide sweeps and passive collections.
 3. Quick-Empty Dock: Swap light passive net bags periodically without vessel downtime.

Smart Municipalities

Integration within city municipal sanitation boards to manage public ponds, drainage outlets, and canals.

Historical Temple Ponds

Perfect non-disruptive, silent cleaning solution for enclosed, historical temple ponds and community pools.

Industrial Sewage

Deployment within sewage channels and holding basins to capture floating debris prior to mechanical filtration.

Private Golf Clubs & Resorts

Maintains clean, aesthetic lake surfaces for lakeside golf clubs, resorts, and private residential communities.

Deployed directly to municipal corporate trusts,
environmental initiatives, and resort chains.
Cleaning-as-a-Service (CaaS): Monthly subscription service including scheduled
maintenance, emptyings, and swarm fleet sweeps.
SaaS Environmental Telemetry: Subscription licensing model for localized water
health mapping, debris volumes, and quality forecasts.

*“EcoFloat is not just a machine.
It is a scalable step toward
restoring life to our water bodies.”*

***Project Initiative: Team EcoFloat
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